

GOVERNMENT OF PUNJAB
DEPARTMENT OF MINES & GEOLOGY

STANDARD OPERATING PROCEDURE

District Survey Report (DSR) Portal User Manual

A comprehensive, step-by-step departmental guide to authentication, project management, data entry, review flow, and final report generation.

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1. INTRODUCTION & WORKFLOW ARCHITECTURE

The District Survey Report (DSR) is a critical statutory document mandatory for environmental clearance of mining leases under the Sustainable Sand Mining Management Guidelines. The DSR Punjab Portal coordinates the workflow of drafting, verifying, and publishing these reports across various administrative levels in Punjab.

STATUTORY RULE

Every district must establish and maintain an updated DSR for sand and minor minerals. This portal establishes full lineage tracking and prevents unauthorized modification of data during review cycles.

The workflow follows a strict sequential lifecycle to ensure audit-ready tracking. Below is the system flow mapping the administrative levels:

SCREENSHOT PLACEHOLDER

Please capture the Workflow Diagram panel on the dashboard showing active stages and review logs.

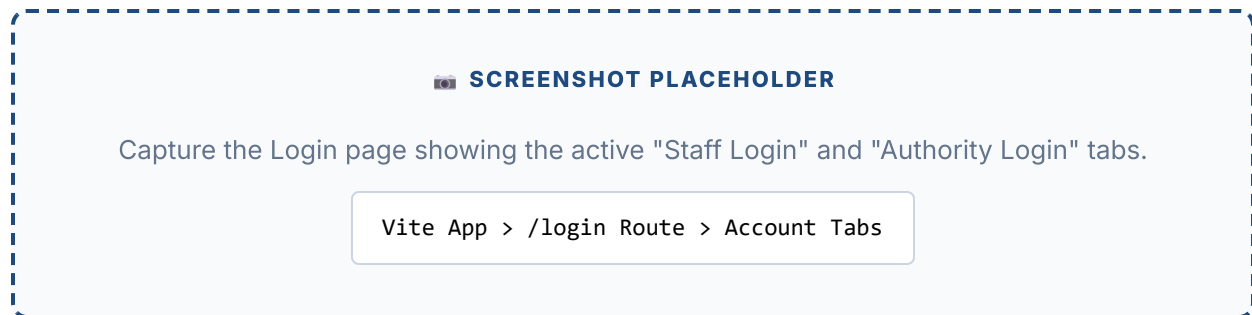
Dashboard > Workflow Overview Panel > Stage Progress Bar

2. PORTAL AUTHENTICATION & USER ROLES

Access control to the portal is aligned with standard departmental designations. Users are categorized under specific roles that grant permissions to edit, review, or approve sections of a District Survey Report.

2.1 Dual Authentication Screen

The portal has two authentication screens designed for security requirements:



A. Department Staff / Geologist Login

Department staff and field geologists use their email address and passwords. Steps to sign in:

1. Verify the **Staff** tab is highlighted.
2. Enter your departmental email (e.g., geologist.asr@punjab.gov.in).
3. Enter your password and click the **Login** button.

B. Authority / Director Login

Senior officials and members of the DSR Committee sign in using their NIC credentials:

1. Click on the **Authority** tab.
2. Input your numeric **NIC ID**.
3. Input your 4-to-6 digit secure **PIN**.
4. Click **Login** to enter.

3. DASHBOARD NAVIGATION & PROJECT SELECTION

Upon login, you are routed to the central workspace dashboard. The header displays active project files, system notices, and quick-access indicators.

SCREENSHOT PLACEHOLDER

Capture the central projects table list showing search parameters and status filters.

Projects Grid > Search Input > Status Filter Tags ("Draft", "Under Review", "Approved")

3.1 Selecting an Existing Project Workspace

To enter a district survey workspace:

1. Go to the **Projects** tab via the top menu.
2. Use the search box to type your target district name (e.g., *Ferozepur*).
3. Click on the row or card containing the project to navigate to the **Project Details Dashboard**.

4. MODULE 1: FRONT MATTER UPLOAD

The Front Matter module is the entry point of the DSR document hierarchy, handling files that establish authenticity.

SCREENSHOT PLACEHOLDER

Capture the Front Matter Editor screen with Cover Page, Compliance Certificate, and Preface file inputs.

Front Matter Panel > File Input Drag-Zones > File Names List

4.1 Upload Instructions

Follow these steps to upload front matter components:

1. Click on **Front Matter** in the project details list.
2. Locate the upload box for the **Cover Page**. Drag your PDF file here.
3. Provide files for the **Certificate of Compliance** and **Preface** similarly.
4. Click the **Save** button to update the state of the document.

5. MODULE 2: CORE CHAPTERS

COMPILATION

DSR Punjab mandates ten structured chapters covering everything from physical geography to mineral potential and environmental safeguards.

SCREENSHOT PLACEHOLDER

Capture the Chapters Editor showing the tabbed list of chapters (1-10) and the rich text editor pane.

Chapters Panel > Left Sidebar Tabs > Content Editor Workspace

5.1 Compiling Chapters

Geologists can draft chapters directly using the online editor or upload prepared documents:

1. Select the chapter from the list (e.g., *Chapter 4: Mineral Wealth of District*).
2. Write or copy-paste text inside the rich text editing box.
3. Use the toolbar to insert tables, headings, and bold text.
4. Set the status to **Completed** and click **Save Chapter**.

6. MODULE 3: PLATES AND MAP ATTACHMENTS

High-resolution maps and geological plates must be attached to the DSR to visually reference mineral deposits and coordinates.

SCREENSHOT PLACEHOLDER

Capture the Plates Upload module displaying uploaded geological map previews and captions.

Plates & Maps Panel > File Preview Grid > Caption Inputs

6.1 Uploading Map Plates

To register a plate:

1. Open **Plates & Maps**.
2. Click the **Add Plate** button.
3. Choose a map file (PDF, PNG, or JPEG format).
4. Enter the Plate details including:
 - **Plate Title** (e.g., *Plate 1 - Drainage Map of District Ludhiana*)
 - **Scale** (e.g., *1:50,000*)
5. Click **Upload Plate** to append.

7. MODULE 4: RIVER CROSS-SECTION DATA

Cross-section profiles help estimate sand extraction depth limits and identify baseline river bed contours.

SCREENSHOT PLACEHOLDER

Capture the Cross Section Profile page with the live elevation profile chart.

Cross Sections Panel > Coordinates Input Grid > SVG Graph Display

7.1 Generating Cross-Section Graphs

To insert profile data:

1. Go to the **Cross Sections** builder page.
2. For each surveyed cross-section point:
 - Input **Chainage Distance** (meters).
 - Input **Elevation Levels** (meters above MSL).
3. The system will automatically plot the data points on the SVG chart.
4. Click **Save Graphs**.

8. MODULE 5: ANNEXURES I-VII & B-K TABLES

Annexures contain tabular database parameters detailing the list of leases, lease durations, sand quantities, and leaseholder names.

SCREENSHOT PLACEHOLDER

Capture the Annexure Editor showing the interactive data grid and the Excel import trigger.

Annexures Panel > Active Table Grid > "Import Excel" Button

8.1 Upload and Edit Instructions

Tables can be populated using the built-in grid tool or spreadsheet template files:

1. Go to the **Annexures** section.
2. Choose the Target Annexure ID (e.g., *Annexure I*).
3. Click ****Import Excel**** and upload your spreadsheet file to fill the table fields automatically.
4. Review the fields for validation errors (highlighted in red).
5. Click **Save Annexure** to commit the data.

9. MODULE 6: REPLENISHMENT CALCULATIONS

Replenishment calculations help determine the sustainability of the mining lease by estimating sand deposit rates after monsoons.

SCREENSHOT PLACEHOLDER

Capture the Replenishment Builder page displaying pre-monsoon and post-monsoon inputs and calculated volumes.

Replenishment Panel > Input Form > Calculations Summary Card

9.1 Inputting Replenishment Parameters

To calculate replenishment rates:

1. Navigate to **Replenishment**.
2. Enter the pre-monsoon sand levels.
3. Enter the post-monsoon sand levels.
4. Click the ****Calculate**** button to calculate the volume variance.
5. Save the parameters to attach this calculation sheet to the final report.

10. MODULE 7: WORKFLOW REVIEW & APPROVAL

Every report must move through the review pipeline to log approvals and observations from inspectors and directors.

SCREENSHOT PLACEHOLDER

Capture the Reviewer workflow interface showing audit trails, comments log, and approval triggers.

Workflow & Review Panel > Comments list > Action Buttons ("Approve", "Return")

10.1 Steps for Reviewing Officers

1. Navigate to the ****Reviewer & Workflow**** section.
2. Inspect each document component listed.
3. If errors are found, click ****Return to Draft**** and write your comment.
4. If the document is correct, select ****Approve Section****.
5. Enter your review details to sign off on the report stage.

11. MODULE 8: FINAL REPORT COMPILATION & PDF GENERATION

The PDF generation engine compiles all approved chapters, annexure grids, front matter attachments, and map plates into a print-ready document.

SCREENSHOT PLACEHOLDER

Capture the Report Preview and PDF Compilation screen with header/footer settings and the download button.

Report Preview Screen > Header/Footer Settings Form > "Download Final PDF" Button

11.1 Generating and Downloading DSR

1. Select ****Report Preview**** or ****Generate PDF**** from the project menu.
2. Review the compiled pages in the live viewer.
3. Configure the header and footer margins and names.
4. Click ****Download Final PDF**** to save the document to your local storage.

IMPORTANT CHECKLIST BEFORE DOWNLOADING

Ensure all sections are marked as "Approved" in the workflow page. Unapproved sections will trigger validation warnings or export with watermarks.